Chemotherapy for soft tissue sarcoma

Chemotherapy uses drugs to treat many different types of cancer. It is most commonly given as an injection into a vein or as tablets or capsules.

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About chemotherapy for soft tissue sarcoma

Chemotherapy uses anti-cancer (cytotoxic) drugs to destroy cancer cells.

Not everyone has chemotherapy. It depends on the risk of soft tissue sarcoma coming back and the type of sarcoma you have. Chemotherapy may not work well for some types of soft tissue sarcoma. Your doctor can tell you whether chemotherapy is a helpful treatment for you.

It may be given:

- after surgery, to try to reduce the risk of the cancer coming back (adjuvant treatment)
- before surgery, to shrink the tumour and make it easier to remove with surgery (neo-adjuvant treatment).

It may also be used to relieve symptoms and help control sarcoma that cannot be completely removed, or has spread to other parts of the body. It may help to improve your quality of life. This is called palliative treatment.

Having chemotherapy

The drugs you have can depend on the type of sarcoma. You usually have a combination of drugs, but some are given on their own. If certain drugs do not work well for you, your doctor may talk to you about trying different ones.

The most commonly used drugs are:

- doxorubicin
- ifosfamide
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liposomal doxorubicin

- gemcitabine
- docetaxel
- paclitaxel
- trabectedin
- dacarbazine
- cyclophosphamide
- vinorelbine.

The chemotherapy drugs are usually given by injection into a vein (intravenously). Some drugs can be given as tablets. You have chemotherapy as a session of treatment. You may have chemotherapy as an outpatient. Or you may need to go into hospital for a few days. Each treatment session is followed by a rest period of a few weeks. This is to allow your body to recover from any side effects.

Chemotherapy and the rest period make up a cycle of your treatment. The number of cycles you have will depend on the type of sarcoma you have and how well it responds to the treatment.

Getting support

Macmillan is here to support you. If you would like to talk, you can:

- Call the Macmillan Support Line for free on 0808 808 00 00.
- Chat to our specialists online.
- Visit our chemotherapy forum to talk with people who have had chemotherapy, share your experience, and ask questions.

Chemotherapy into a limb

You can sometimes have chemotherapy directly into an arm or leg. It can help control sarcoma that cannot be removed with surgery or has come back in one area. This is called isolated limb perfusion.

Having the drugs directly into the affected limb means that very high doses can be given. The limb is isolated using a tight band. This is called a tourniquet. It stops the chemotherapy drugs going to other areas of the body. You are unlikely to get common chemotherapy side effects, such as feeling sick. The affected limb may swell and the skin may turn red.

Side effects of chemotherapy for soft tissue sarcoma

Chemotherapy drugs cause side effects. But there are usually ways these can be controlled. Your doctor or nurse will give you information about the likely side effects of your treatment. They will also tell you what can be done to control and manage side effects.

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Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an infection. A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

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- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Feeling sick

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

If you feel sick, take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

Sore mouth and throat

This treatment may cause a sore mouth and throat. You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

If your mouth or throat is sore:

- tell your nurse or doctor – they can give you a mouthwash or medicines to help
- try to drink plenty of fluids
- avoid alcohol, tobacco, and foods that irritate your mouth and throat.

Hair loss

Your hair will get thinner. Or you may lose all the hair from your head. You may also lose your eyelashes and eyebrows, as well as other body hair. Hair loss usually starts after your first or second treatment.

Scalp cooling is a way of lowering the temperature of your scalp to help reduce hair loss. Your nurse can tell you whether this is an option for you.

If you want to cover up hair loss, there are different ways to do this. Your nurse will give you information about coping with hair loss.

Remember to protect your skin from the sun. Use suncream with a sun protection factor (SPF) of at least 30 on your scalp. Or cover up with a hat or scarf.

Hair loss is almost always temporary. Your hair will usually grow back after treatment ends.

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Feeling tired

Feeling tired is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Effects on the kidneys

This treatment can affect how the kidneys work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called peripheral neuropathy. You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Effects on the heart

This treatment can affect how the heart works. You may have tests to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing heart problems, your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
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difficulty breathing.

Fertility

Some cancer drugs can affect whether you can get pregnant or make someone pregnant. If you are worried about this, it is important to talk with your doctor before you start treatment.

It may be possible to store sperm, eggs or embryos for use in the future.

Early menopause

Sometimes chemotherapy treatment causes an early menopause, and they may have symptoms such as hot flushes and sweats. Hormone replacement therapy (HRT) can often be given to replace the hormones that are no longer being produced. People with a gynaecological sarcoma may not be able to have HRT, because the cancer may be sensitive to hormones.

You may find it helpful to talk about all this with your doctor or a support organisation. You can also talk to one of our cancer support specialists for free on 0808 808 00 00.

Booklets and resources

Booklets

01 Mar 2022

Side effects from chemotherapy - easy read

About our information

References

Below is a sample of the sources used in our soft tissue sarcoma information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Gronchi A, Miah AB et al. Soft tissue and visceral sarcomas: ESMO-EURACAN-GENTURIS Clinical practice guidelines for diagnosis, treatment and follow-up. Annals of Oncology, 2021; 32, 11, 1348-1365 [accessed May 2022].

Casali PG, Blay JY et al. Gastrointestinal stromal tumours: ESMO-EURACAN-GENTURIS Clinical practice guidelines for diagnosis, treatment and follow-up. Annals of Oncology, 2022; 33,1, 20-33 [accessed May 2022].

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Reviewers

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by senior medical editor Fiona Cowie, Consultant Clinical Oncologist.

Our cancer information has been awarded the PIF TICK. Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

Date reviewed

Reviewed: 01 August 2022 | Next review: 01 August 2025

This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

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How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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